

Chapter 28: Mammals

Origin and Evolution of Mammals

- The evolutionary descent of mammals from their earliest amniote ancestors is well documented.
- Skull structures and teeth are the most abundant fossils used to identify evolutionary descent.
- Synapsids: The group including mammals and their ancestors, characterized by a pair of temporal openings in the skull for jaw muscle attachment.
 - Synapsids were the first amniote group to radiate widely into terrestrial habitats.
 - Pelycosaurs: Early synapsids, resembling lizards but not closely related. They were the most common and largest amniotes in the early Permian.
 - Therapsids: Arose from early carnivorous synapsids. They had an efficient erect gait with upright limbs positioned beneath the body.
 - Cynodonts: A group of therapsids that survived into the Mesozoic. They evolved features supporting a high metabolic rate:
 - Specialized jaw musculature for a stronger bite.
 - Heterodont teeth for better food processing.
 - Turbinate bones in the nasal cavity to aid heat retention.
 - Secondary bony palate enabling breathing while eating.
 - Loss of lumbar ribs, correlated with the evolution of a diaphragm.

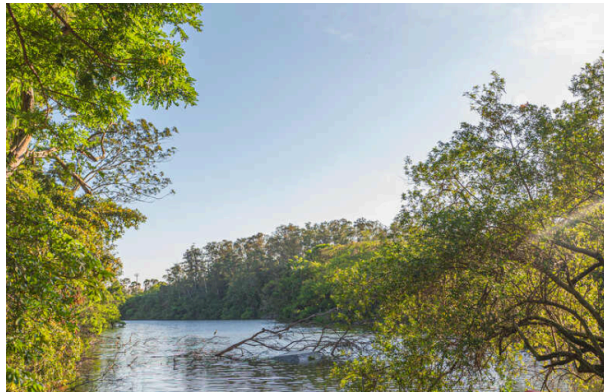


Figure 1: Skulls of early amniotes, showing the pattern of temporal openings that distinguish the three groups: Anapsid, Synapsid, and Diapsid.



Figure 2: Evolution of major groups of synapsids. The synapsid lineage began with pelycosaurs, continued with therapsids and cynodonts, leading to modern mammals (monotremes, marsupials, and placentals).

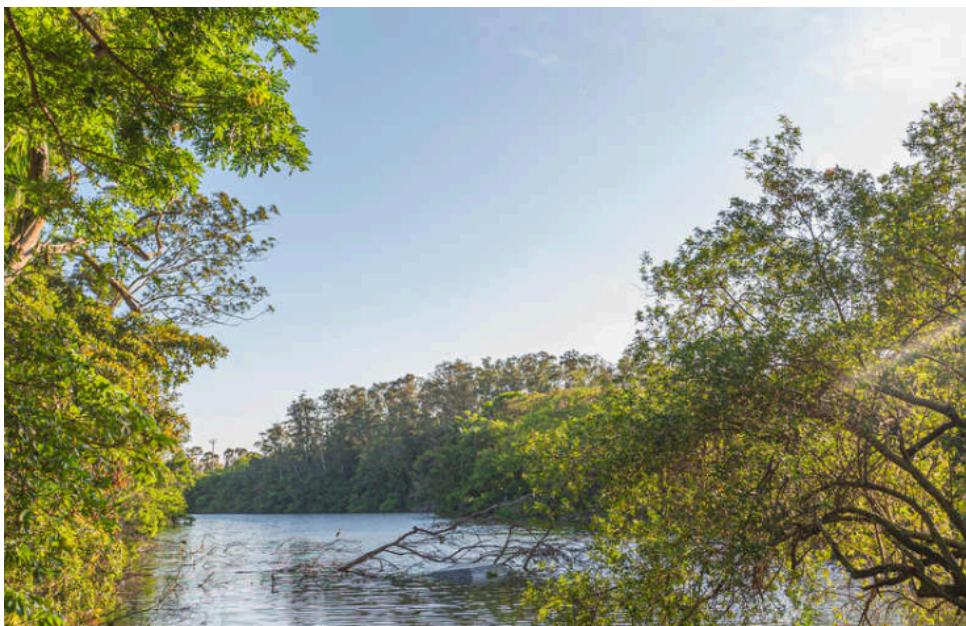


Figure 3: Abbreviated cladogram of synapsids emphasizing origins of important characteristics of mammals, such as the enlarged dentary bone and heterodont teeth.

- Earliest Mammals: Appeared in the late Triassic.
 - Small, mouse- or shrew-sized.
 - Diphyodont dentition: Teeth replaced only once (deciduous and permanent sets).
 - Middle Ear Bones: Malleus and incus originated from the articular and quadrate bones (previously the jaw joint). The stapes was already present.
 - Dentary-squamosal joint: The new jaw joint defining fossil mammals.
 - Endothermic, with hair for insulation and sebaceous/sweat glands.
 - Young likely hatched from eggs and were dependent on maternal milk.



Figure 4: Evolution of the jaw joint and middle ear bones in the synapsid lineage. The transition from the articular-quadrato jaw joint to the dentary-squamosal joint, with the former bones becoming the malleus and incus of the middle ear.

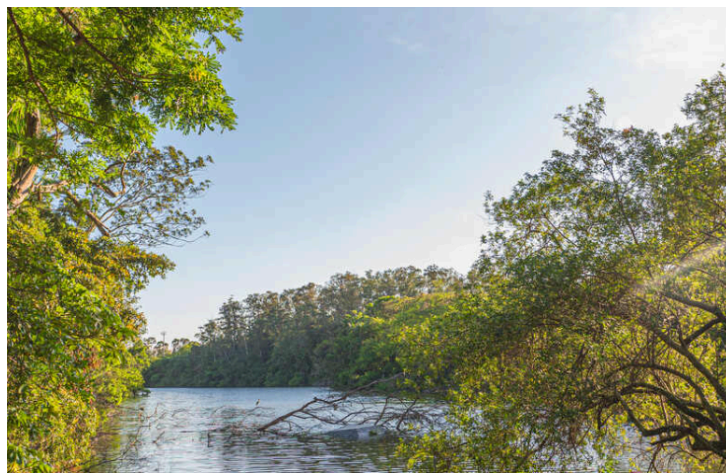


Figure 5: Sagittal section of the head of a rabbit, showing the secondary palate, turbinate bones, and other mammalian skull features.

Structural and Functional Adaptations of Mammals

Integument and Its Derivatives

- Mammalian skin is thicker than in other vertebrates, composed of epidermis and dermis.
- Hair: Characteristic of mammals.
 - Grows from a hair follicle sunk into the dermis.
 - Composed of dead, keratin-packed epidermal cells.
 - Pelage (fur coat) consists of:
 - Underhair: Dense and soft for insulation.
 - Guard hair: Coarse and longer for protection and coloration.
 - Molts: Periodic shedding of the entire coat (usually twice a year).
 - Vibrissae (“whiskers”): Sensory hairs providing tactile sense.

- Quills: Modified hairs for defense (e.g., porcupines).



Figure 6: Structure of human skin (epidermis and dermis) and hypodermis, showing hair follicles, sebaceous glands, sweat glands, and arrector pili muscles.



Figure 7: American beaver, **Castor canadensis**, gnawing on an aspen tree. Note the thick waterproof pelage.

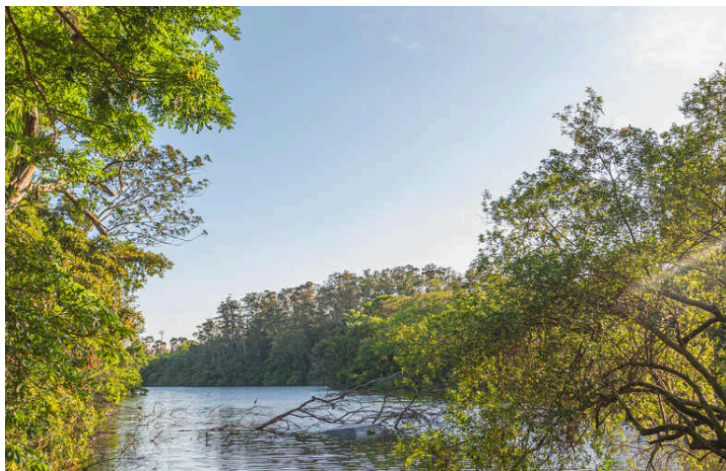


Figure 8: Snowshoe, or varying, hare, **Lepus americanus**. A, Brown summer coat. B, White winter coat for camouflage in snow.



Figure 9: Dogs are frequent victims of a porcupine's impressive quills, which are modified hairs used for defense.

Horns and Antlers

- True Horns (Family Bovidae): Hollow sheaths of keratinized epidermis around a core of bone. Not shed, not branched, grow continuously, found in both sexes.
- Antlers (Family Cervidae): Branched and composed of solid bone when mature.
 - Develop beneath vascular soft skin called velvet.
 - Shed after the breeding season.
 - Usually only in males (except caribou).
- Rhinoceros Horn: Hairlike keratinized filaments cemented together, not attached to the skull.

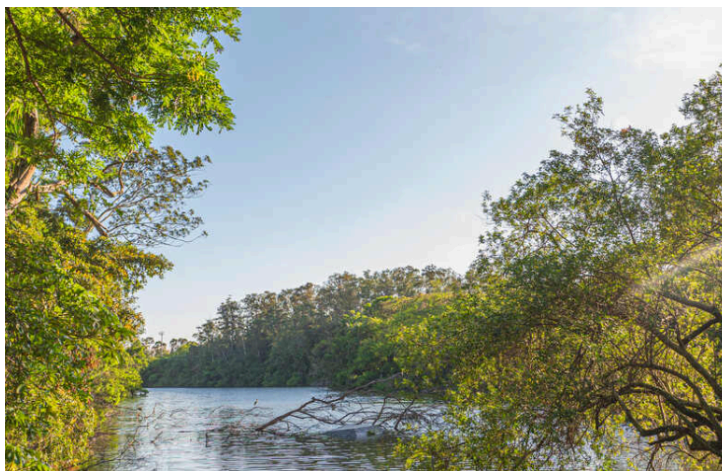


Figure 10: Annual growth of buck deer antlers. A, Growth begins in spring. B, Rapid bone growth. C, Velvet sheds. D, Antlers shed after breeding season.

Glands

- Sweat Glands: Tubular, coiled glands for thermoregulation.
 - Eccrine glands: Secrete watery fluid for cooling (evaporation).
 - Apocrine glands: Larger, secrete milky fluid, correlated with reproductive function (not heat regulation).
- Scent Glands: Used for communication, marking territory, warning, or defense.
- Sebaceous Glands: Associated with hair follicles, secrete sebum (fatty mixture) to keep skin and hair pliable.

- Mammary Glands: Modified apocrine glands for milk secretion to nourish young. Occur on all female mammals.

Food and Feeding

- Teeth: Reveal life habits.
 - Heterodont: Differentiated into incisors, canines, premolars, and molars.
 - Incisors: Snipping or biting.
 - Canines: Piercing.
 - Premolars and Molars: Shearing, slicing, crushing, or grinding.
 - Diphyodont: Two sets of teeth (deciduous and permanent).
- Feeding Specializations:
 - Insectivores: Small, sharp-pointed teeth for puncturing exoskeletons. Short intestinal tract.
 - Herbivores: Browsers, grazers, and gnawers.
 - Canines absent or reduced; molars broad and high-crowned.
 - Gnawers (Rodents): Chisel-sharp incisors that grow continuously.
 - Digestion of cellulose requires anaerobic bacteria and protozoa in fermentation chambers.
 - Cecum: Fermentation chamber in some herbivores (horses, rabbits).
 - Ruminants: Four-chambered stomach (rumen, reticulum, omasum, abomasum) for efficient fermentation and nutrient extraction.
 - Carnivores: Biting and piercing teeth, powerful claws. Shorter digestive tract, small or absent cecum. Discrete meals.
 - Omnivores: Use both plants and animals. Versatile dentition.

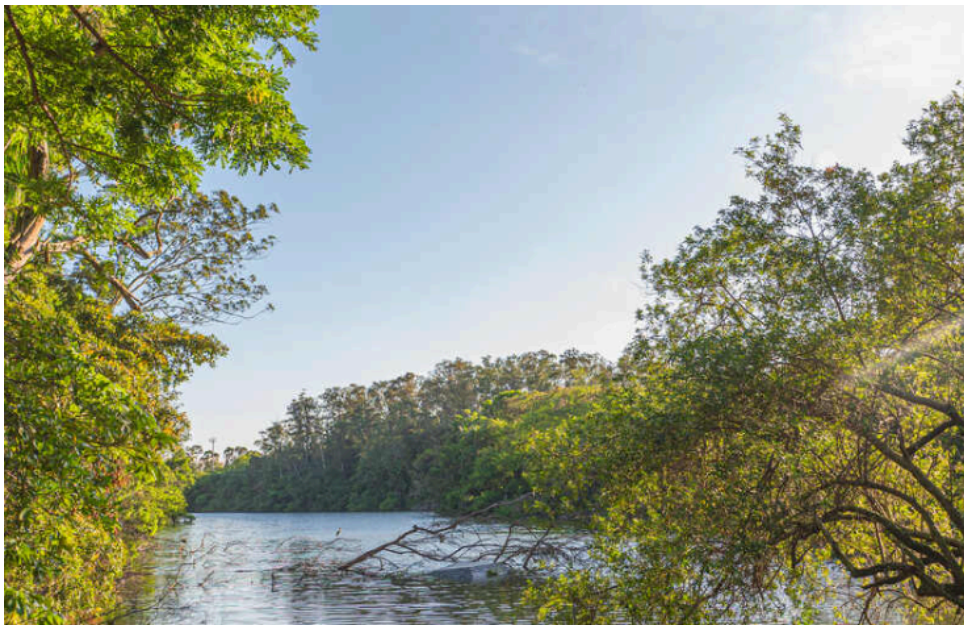


Figure 11: Feeding specializations of major trophic groups of eutherian mammals, showing dental adaptations for insectivores, herbivores, carnivores, and omnivores.



Figure 12: Digestive systems of mammals, showing different morphology with different diets: Insectivore, Nonruminant Herbivore, Ruminant Herbivore, and Carnivore.

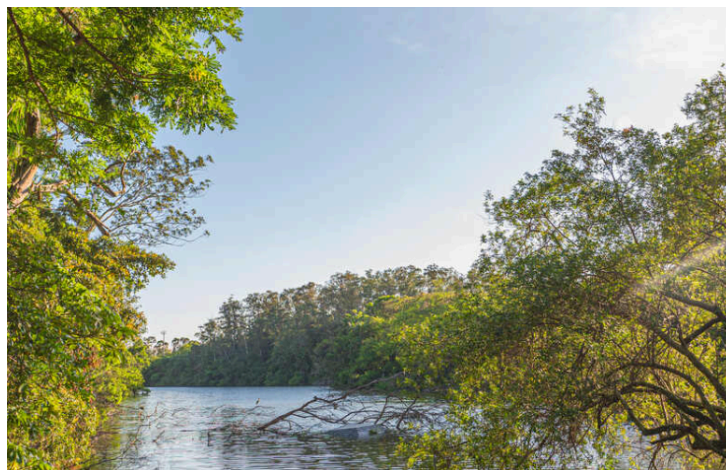


Figure 13: Lionesses, **Panthera leo**, eating a wildebeest. Carnivores have specialized teeth and claws for killing and processing prey.

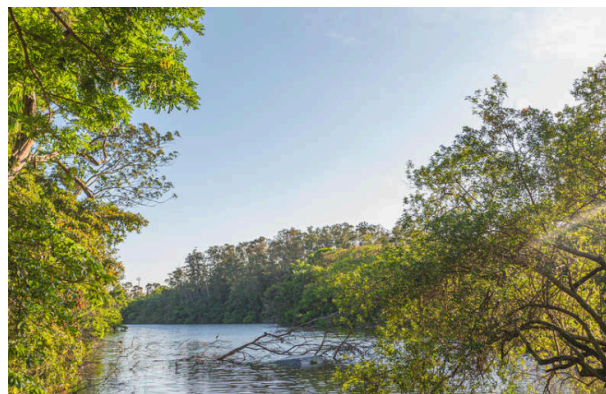


Figure 14: Eastern chipmunk, **Tamias striatus**, with cheek pouches stuffed with seeds to be carried to a hidden cache.

Body Weight and Food Consumption

- Metabolic rate is inversely proportional to body size.
- Small mammals have a much higher metabolic rate per gram of body weight than large mammals.
- Small mammals must consume much more food relative to their body weight than large mammals.



Figure 15: Relationship between body weight and metabolic rate for mammals. Smaller mammals have higher metabolic rates per gram of body weight than larger mammals.

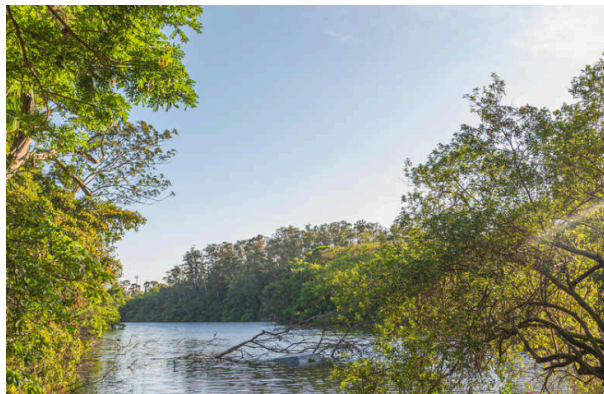


Figure 16: Shorttail shrew, **Blarina brevicauda**, eating a grasshopper. Shrews have high metabolic rates and voracious appetites.

Migration

- Few terrestrial mammals migrate due to the energy cost of locomotion.
- Caribou: Mass migrations between winter taiga ranges and summer tundra calving ranges.
- Whales: Longest mammalian migrations (e.g., Gray whales).
- Fur Seals: Migrate between breeding grounds (e.g., Pribilof Islands) and wintering grounds.
- Bats: Some species migrate to warmer climates for winter.



Figure 17: Barren-ground caribou, **Rangifer tarandus**, of Canada and Alaska. A, Adult male.
B, Map showing summer and winter ranges and migration routes.

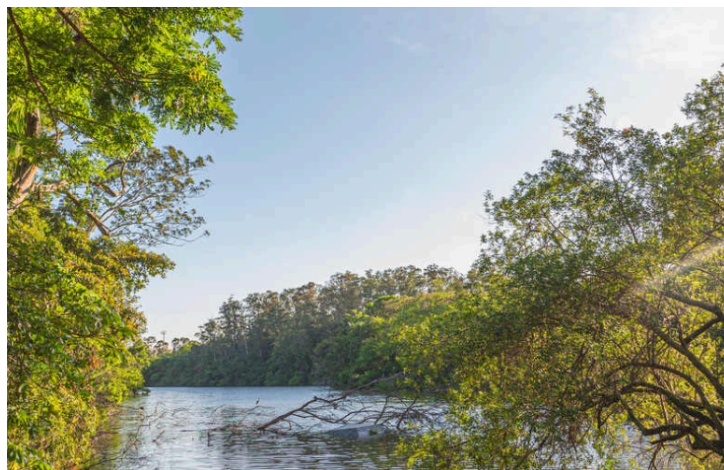


Figure 18: Annual migrations of northern fur seals, **Callorhinus ursinus**, showing the separate wintering grounds of males and females and their convergence on the Pribilof Islands.

Flight and Echolocation

- Flight: Evolved in bats (Order Chiroptera).
 - Wings are modified forelimbs with elongated digits supporting a membrane.
 - Gliding: Evolved in flying squirrels, lemurs, etc., using a patagium.
- Echolocation: Used by bats and cetaceans for navigation and feeding.
 - Bats emit ultrasonic pulses (frequency modulated).
 - Echoes provide information on distance, size, texture, and movement of objects.
 - Cetaceans: Toothed whales use echolocation (clicks produced in nasal passages, focused by the melon).



Figure 19: Northern flying squirrel, **Glaucomys sabrinus**, gliding. The patagium extends between the limbs to allow gliding.



Figure 20: Echolocation of an insect by a little brown bat, **Myotis lucifugus**. The bat emits frequency-modulated pulses that increase in rate as it approaches prey.

Reproduction

Reproductive Cycles

- Estrous Cycle: Females are receptive to males only during estrus (heat).
 - Monestrous: Single estrus during breeding season (e.g., dogs, bats).
 - Polyestrous: Recurrence of estrus during breeding season (e.g., mice, squirrels).
- Menstrual Cycle: In Old World monkeys and humans, the endometrium is shed (menstruation) if fertilization does not occur.



Figure 21: African lions, **Panthera leo**, mating. Lions have a defined estrous cycle.

Reproductive Patterns

- Monotremes (Egg-laying):
 - Oviparous.
 - Eggs incubated in a burrow or pouch.
 - Young nourished by milk lapped from the mother's belly (no nipples).
 - Example: Duck-billed platypus, Echidna.
- Marsupials (Pouched):
 - Viviparous with a choriovitelline (yolk sac) placenta.
 - Brief gestation; young born at an embryonic stage.
 - Prolonged lactation and parental care in a pouch (marsupium).
 - Embryonic Diapause: Arrested development of an embryo in the uterus while a young is suckling in the pouch (e.g., Kangaroos).
- Placental Mammals (Eutherians):
 - Viviparous with a chorioallantoic placenta.
 - Prolonged gestation; young born at a more advanced stage.
 - Investment in gestation rather than lactation.



Figure 22: Comparison of gestation and lactation periods between matched pairs of ecologically similar species of marsupial and placental mammals. Marsupials have short gestation and long lactation; placentals have long gestation and shorter lactation.

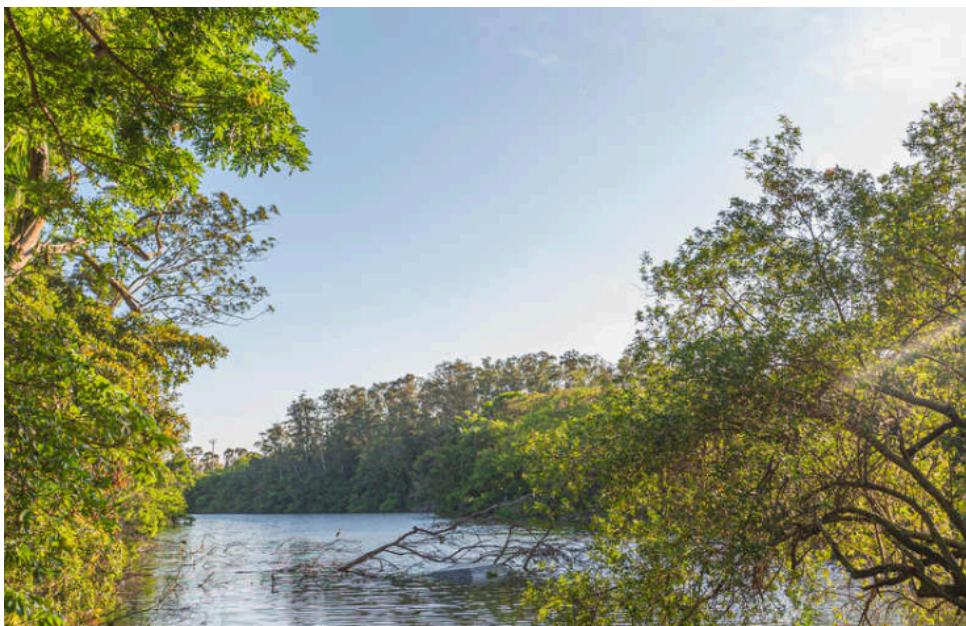


Figure 23: Kangaroos have a complicated reproductive pattern involving embryonic diapause, allowing a mother to have three young in different stages of development simultaneously.



Figure 24: Southern opossums, **Didelphis marsupialis**, 15 days old, fastened to teats in mother's pouch. They are born at a very early stage of development.

Territory and Home Range

- Territory: Area defended against individuals of the same species.
 - Marked with scent glands, urine, or feces.
 - Size varies with animal size and feeding habits.
- Home Range: Larger foraging area surrounding the territory; not defended and may overlap with others.
- Social Behavior:
 - Beavers: Monogamous, defend colonies.
 - Prairie Dogs: Live in large social "towns" subdivided into family units.

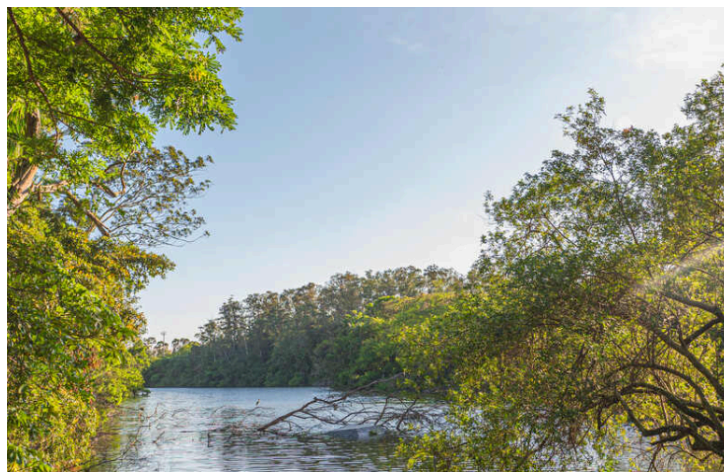


Figure 25: Diagram of a beaver lodge and dam. Beavers create their own aquatic environment and defend their territory.



Figure 26: Immature black-tailed prairie dogs, **Cynomys ludovicianus**, greeting an adult. They live in large social communities called towns.

Mammalian Populations

- Populations fluctuate due to density-independent (climate, catastrophes) and density-dependent (crowding, competition) factors.
- Cycles of Abundance:
 - Lemmings: Mass migrations and population crashes every 3-4 years.
 - Snowshoe Hares: 10-year cycles correlated with lynx (predator) populations. Crashes may be due to “shock disease” (endocrine imbalance) from crowding.

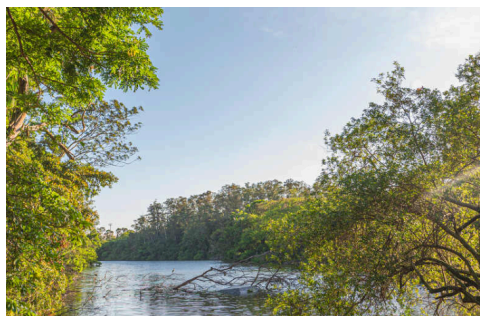


Figure 27: Collared lemming, **Dicrostonyx** sp. Populations of lemmings fluctuate widely in cycles.



Figure 28: Changes in population size of varying hare and lynx in Canada over a 90-year period, showing the classic predator-prey cycle.

Humans and Mammals

- Domestication: Dogs, cats, cattle, sheep, pigs, horses, etc.
- Impact:
 - Rodents as pests and disease carriers (bubonic plague, typhus).
 - Overexploitation (e.g., whales).
 - Habitat destruction.

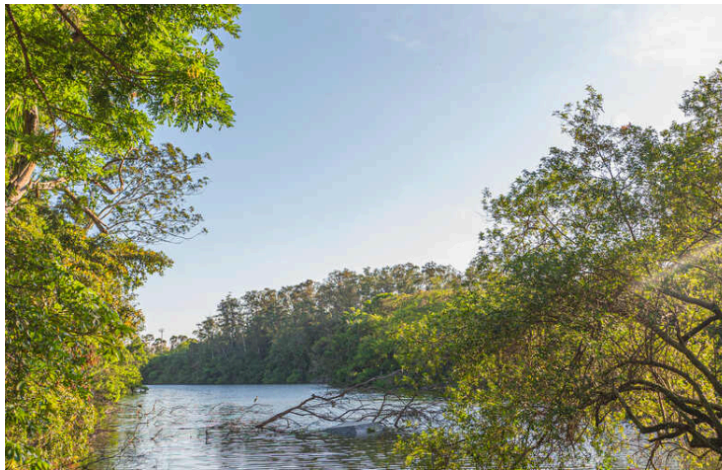


Figure 29: Herd of reindeer, **Rangifer tarandus**, during annual roundup by Laplanders. Reindeer are semi-domesticated caribou.



Figure 30: Brown rat, **Rattus norvegicus**, a successful commensal with humans that causes damage and spreads disease.

Human Evolution

- Humans are primates.
- Primate Characteristics: Grasping fingers, flat fingernails, forward-pointing eyes (binocular vision), large cerebral cortex.
- Primate Lineages:
 - Strepsirhini: Lemurs, lorises (wet noses).
 - Haplorhini: Tarsiers, monkeys, apes, humans (dry noses).
 - New World Monkeys: Prehensile tails, flat noses.
 - Old World Monkeys: Close-set nostrils, no prehensile tails.
 - Apes (Hominoids): Larger cerebrum, no tail.
- Hominids: Humans, orangutans, gorillas, chimpanzees.

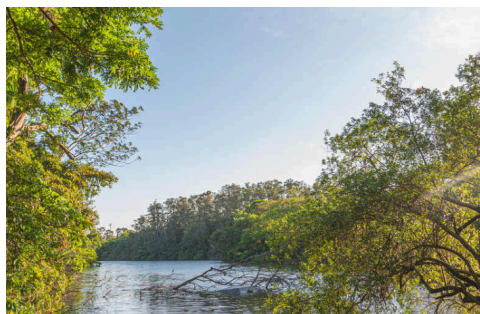


Figure 31: A prosimian, the Mindanao tarsier, **Tarsius syrichta carbonarius**. Tarsiers are small, nocturnal primates with large eyes.



Figure 32: Monkeys. A, Red howler monkeys (New World). B, Olive baboons (Old World).

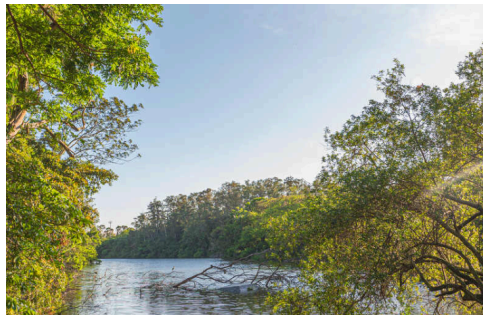


Figure 33: Gorillas, **Gorilla gorilla**, examples of apes (Hominidae).

- Early Humans:
 - **Sahelanthropus tchadensis**: 7 million years ago (oldest known).
 - **Ardipithecus ramidus**: 4.4 million years ago.
 - **Australopithecus afarensis** ("Lucy"): 3.2 million years ago, bipedal.
 - **Homo habilis**: 2.4-1.8 million years ago, used stone tools.
 - **Homo erectus**: 1.8 million years ago, migrated out of Africa, used fire.
 - **Homo neanderthalensis**: 150,000-30,000 years ago, robust, cold-adapted.
 - **Homo sapiens**: Arose in Africa 200,000 years ago.



Figure 34: Hominid skulls, showing several of the best-known lines preceding modern humans, including **Sahelanthropus**, **Australopithecus**, and early **Homo** species.

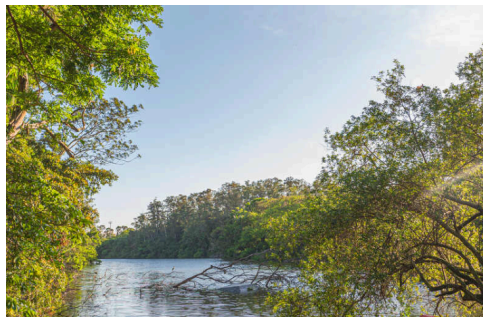


Figure 35: Lucy (**Australopithecus afarensis**), the most nearly complete skeleton of an early human ever found.

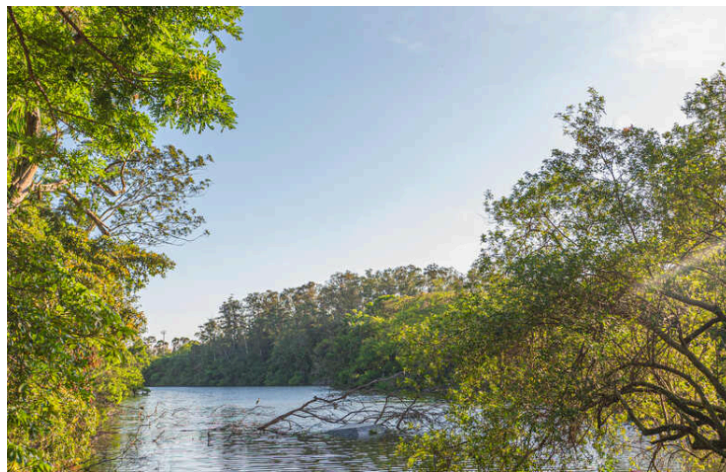


Figure 36: A reconstruction of the appearance of Lucy (right) compared with a modern human (left).

Classification of Living Mammalian Orders

- Subclass Prototheria:

- Order Monotremata: Egg-laying mammals (Platypus, Echidna).
- Subclass Theria:
 - Infraclass Metatheria (Marsupials):
 - Order Didelphimorphia: Opossums.
 - Order Diprotodontia: Kangaroos, koalas, wombats.
 - And others (Paucituberculata, Microbiotheria, Dasyuromorphia, Peramelemorphia, Notoryctemorphia).
 - Infraclass Eutheria (Placentals):
 - Order Insectivora: Shrews, moles, hedgehogs.
 - Order Chiroptera: Bats.
 - Order Primates: Prosimians, monkeys, apes, humans.
 - Order Xenarthra: Anteaters, armadillos, sloths.
 - Order Lagomorpha: Rabbits, hares, pikas.
 - Order Rodentia: Squirrels, rats, beavers, porcupines (largest order).
 - Order Carnivora: Dogs, cats, bears, seals, walruses.
 - Order Proboscidea: Elephants.
 - Order Perissodactyla: Odd-toed ungulates (horses, rhinos).
 - Order Artiodactyla: Even-toed ungulates (cattle, deer, pigs, camels, whales - Cetacea is now nested here or sister).
 - Order Cetacea: Whales, dolphins (often grouped with Artiodactyla as Cetartiodactyla).
 - And others (Macroscelidea, Dermoptera, Scandentia, Pholidota, Tubulidentata, Hyracoidea, Sirenia).

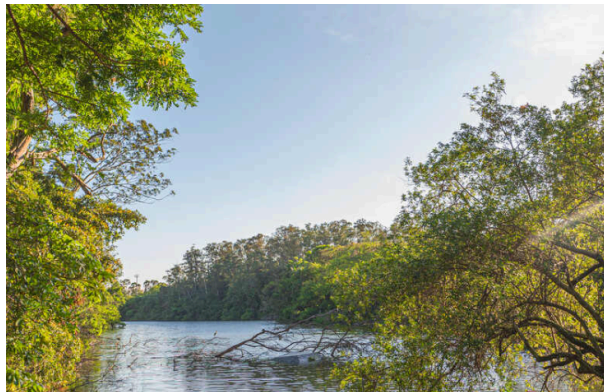


Figure 37: Nine-banded armadillo, **Dasyus novemcinctus**. Order Xenarthra.

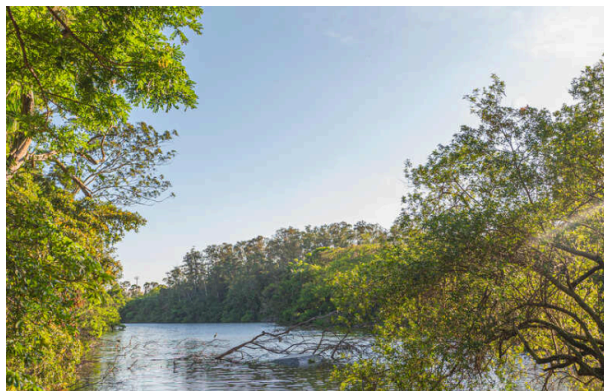


Figure 38: A pika, **Ochotona princeps**, atop a rockslide. Order Lagomorpha.



Figure 39: Grizzly bear, ***Ursus arctos horribilis***. Order Carnivora, family Ursidae.

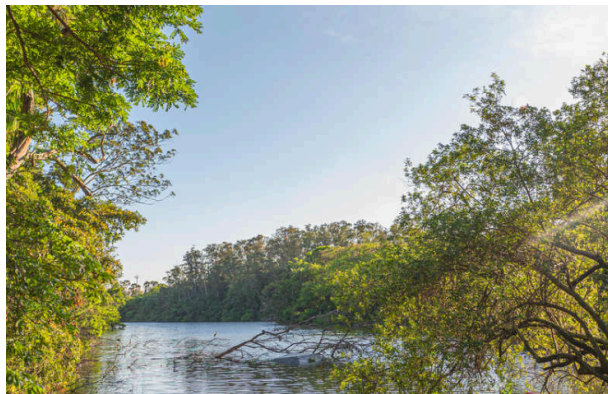


Figure 40: A Galápagos sea lion bull, ***Zalophus californianus***. Order Carnivora, family Otariidae.

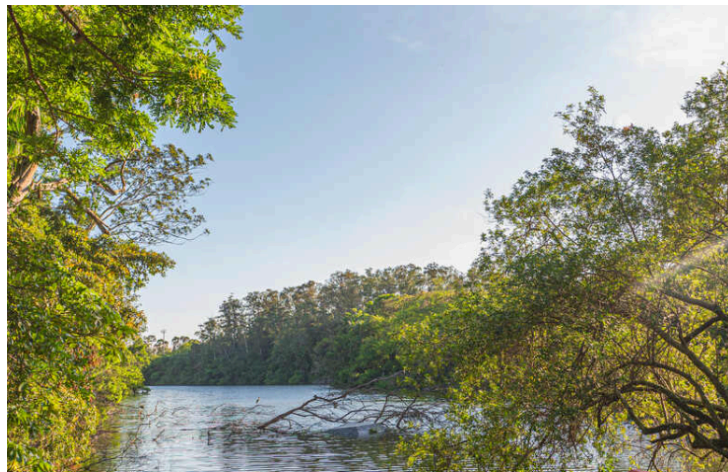


Figure 41: Odd-toed (Perissodactyla) and even-toed (Artiodactyla) ungulates. Comparison of foot structure.



Figure 42: Humpback whale, **Megaptera novaeangliae**, breaching. Order Cetacea.